All implicit frames are orthonormal and right oriented.

Setup 1: Consider the hyperbola

$$\mathcal{H}_a : \frac{x^2}{a^2} - \frac{y^2}{4} = 1 \quad \text{where} \quad a > 0.$$

1. (1 point) Determine the value of a for which the asymptotes have an angle of $45°$ with the x-axis.
2. (1 point) For $a = 1$, determine the equations of the tangent lines to $\mathcal{H}_a$ that pass through the point $Q(0,1)$.
3. (1 point) Determine the values of a for which $\mathcal{H}_a$ is tangent to $\ell : x - y = 1$.

Setup 2: Consider the hyperbola

$$\mathcal{H} : x^2 - y^2 = 1.$$

4. (1 point) Determine the distance between the focal points of the hyperbola $\mathcal{H}$.
5. (1 point) Determine the tangent lines to $\mathcal{H}$ that are parallel to the line $\ell : 5x - 4y - 1 = 0$.
6. (1 point) Give the equation for the hyperbola obtained by rotating $\mathcal{H}$ counterclockwise by $45°$.
7. (1 point) Determine the values of λ for which the point $P(5, \lambda)$ does not lie on any tangent line of the hyperbola $\mathcal{H}$.

Setup 3: Consider the parabola

$$\mathcal{P} : y^2 = 2x.$$

8. (1 point) Give an equation for the directrix of $\mathcal{P}$.
9. (1 point) Determine the tangent lines to $\mathcal{P}$ that have a $45°$ angle with the x-axis.
10. (1 point) Let α be one of the angles formed by the x-axis and the tangent line to $\mathcal{P}$ at the point $P(2,-2) \in \mathcal{P}$. Determine the possible values for $\cos\alpha$.
11. (1 point) Determine the values of a for which the line $\ell_a : x - ay + 2 = 0$ is tangent to $\mathcal{P}$.

Answers:

1. 2
2. $\sqrt{5}x - y + 1 = 0$ and $\sqrt{5}x + y - 1 = 0$
3. $\sqrt{5}$
4. $2\sqrt{2}$
5. $5x - 4y - 3 = 0$ and $5x - 4y + 3 = 0$
6. $2xy - 1 = 0$
7. $-2\sqrt{6} < \lambda < 2\sqrt{6}$
8. $x = -\frac{1}{2}$
9. $2x - 2y + 1 = 0$ and $2x + 2y + 1 = 0$
10. $\pm\frac{2}{\sqrt{5}}$
11. ± 2